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Rethinking Uncertainty Quantification in Diffusion-Bridge Medical Image

Translation: Pitfalls, Baselines, and Boundaries (system: TrustBridge — a training-free trust layer with Convergence Residual Fields)

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Code and weights: https://github.com/ASHDUSK/TrustBridge-

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ABSTRACT

Two pitfalls quietly corrupt most uncertainty-quantification (UQ) claims made for generative medical image translation. First, *background air pixels*: they are trusted and near-zero-error, so pooling them with body pixels inflates whole-image UQ metrics severalfold—we show a headline localization score of 0.93 collapsing to 0.78 under a within-body protocol, and retract our own earlier numbers that benefited from it. Second, *dispersion is not error*: across an equal-budget comparison with TTA and input-noise baselines, every dispersion-style signal—including ours—is at chance on T2→T1, where residual error is dominated by fine texture. We then ask what *does* survive. From the self-consistent recursive diffusion-bridge sampler (SelfRDB) we harvest two training-free signals: the *Convergence Residual Field* (CReF), the per-pixel discrepancy between successive recursive target estimates that the original sampler computes and discards, and *stochastic-bridge ensemble uncertainty*. Fused into a pixel-wise trust map with a three-tier QC gate and a six-class input gate with out-of-distribution rejection, they deliver, under the honest protocol: AUROC 0.90 (T1→T2, self-trained; 0.88 for a single zero-cost chain), 0.91 on real tumor patients, and 0.95 *cross-institution* on never-seen subjects. CReF-triggered routing recovers most of full-budget TTA quality at a fraction of its cost. A dedicated reproducibility analysis explains why officially released weights collapse to $\sim$15 dB on independently prepared data (paper value 31.56 dB)—structurally, not just photometrically (NCC 0.41)—and shows that the trust layer *fails to notice*: the collapsed model receives PASS with $94/100$ mean trust. Self-consistency cannot certify correctness; external validation remains irreplaceable. We release all negative results—the protocol trap, the texture-dominated failure boundary, the absence of tumor localization, the miscalibration of image-level gates—as a boundary map for

the field, and recommend UQ claims in image translation be evaluated within the body, against cheap baselines, and per (model, direction) pair.

1. Introduction

Multi-contrast magnetic resonance imaging (MRI) captures complementary tissue contrast, but acquiring every clinically relevant weighting is expensive, slow, and sometimes contraindicated. Medical image translation—synthesizing an unacquired modality from an acquired one—addresses this scarcity and now achieves excellent fidelity. Denoising diffusion probabilistic models (DDPMs)[1,2] replaced GANs in this task for their training stability, and diffusion bridges connect the source and target distributions directly rather than through an isotropic-noise bottleneck. In particular, SelfRDB[3] introduced a self-consistent recursive diffusion bridge that reaches state-of-the-art multi-contrast translation. Trust, however, has not kept pace with fidelity. Because translation is ill-posed, generative models can *hallucinate*: synthesize plausible structure that is absent from the underlying anatomy. In clinical use, a radiologist cannot distinguish such fabrication from faithful synthesis, and this trust bottleneck—not image quality—is the primary obstacle to deployment. The SelfRDB authors state this gap explicitly: the uncertainty of diffusion bridges "is warranted to [be] evaluate[d]" in future work [3]. We close this gap with an observation: the self-consistent recursive sampler already computes, at every step and every pixel, the evidence needed to assess its own reliability—it simply discards it. SelfRDB reduces the discrepancy between successive recursive estimates to a single stopping scalar; the per-pixel residual field carries the spatial distribution of the model's *inability to self-consist*. Likewise, the bridge endpoint noise and generator latent variables make repeated runs differ; their dispersion marks where the model holds no stable belief. TrustBridge systematizes both signals into a trust map, an image-level Trust Score, and an archivable QC report, with three properties that distinguish it from prior uncertainty machinery: (i) *training-free*—no model change, no labels, no extra network; (ii) *sampler-native*—signals are extracted from quantities the sampler already computes; and (iii) *pixel-level*—unlike image-level hallucination metrics[4], the output supports a radiologist's local reading. A further contribution is methodological. In the course of validation we found that *background air pixels silently inflate whole-image trust metrics*: air is trusted (uniform, easy) and error-free, so pooling it with body pixels flatters any "low trust predicts high error" score. Several numbers we initially reported (a tumor-localization AUROC of 0.87, decile separations of 35–$73\times$) were artifacts of this pooling and do not survive within-body evaluation. All trust metrics in this paper are therefore computed inside a body mask, and we document the inflation effect itself as a caution for future uncertainty evaluations (§§4).

Contributions.

1. We propose the Convergence Residual Field (CReF) and stochastic-bridge ensemble uncertainty as complementary, training-free trust signals for diffusion-bridge translation, fused into a pixel-wise trust map with a three-tier QC gate, plus a six-class input gate with Mahalanobis out-of-distribution rejection that hard-blocks mismatched or anomalous inputs before synthesis (§§3).
2. We introduce a body-masked evaluation protocol for pixel-wise trust metrics, quantify the whole-image inflation it corrects, and re-evaluate our signals against equal-budget dispersion baselines (TTA, input-noise): the fused signal localizes the top-error decile at AUROC $0.90/0.82$ (self-trained T1→T2/T2→T1), beating all equal-budget baselines, 0.91 on real tumor patients, and 0.95 cross-institution; on the *official* weights the same protocol flips the ranking and blinds T2→T1 entirely (§§4).
3. We report seven controlled experiments whose outcomes are mixed or negative—adaptive resampling, learnable aggregation, image-level regression, intensity harmonization, round-trip consistency, the official-weights T2→T1 signal failure, and the semantic-proxy null result—and use them to delineate the method's boundary: dispersion signals track error only where error is spatially structured, and learning-based aggregation overfits six-subject validation sets (§§4, §§5).

2. Related Work

2.1 Medical Image Translation

Unpaired cross-modality synthesis has evolved from adversarial translation —CycleGAN[5] and pix2pix[6] established the unpaired and paired regimes, followed by multi-contrast MRI specialists such as conditional GANs[7] and hybrid CNN–transformers (ResViT[8])—to diffusion-based models. SynDiff [9] introduced adversarial diffusion with a cyclic consistency prior for unsupervised medical translation, and I2I-Galip[10] replaced the discriminator with a vision–language model for efficiency. In parallel, bridges connect the two image distributions directly rather than through an isotropic-noise bottleneck: DDPMs[1] and their score-based generalization[11] provide the mathematical backbone, with score-based NCSN[12], improved schedules[2], direct diffusion bridges with data consistency[13], dual implicit bridges[14], neural Schrödinger bridges[15], and direct iterative inversion[16] spanning the design space. SelfRDB[3] combines a soft-prior bridge with self-consistent recursive sampling and reports state-of-the-art multi-contrast translation, while explicitly leaving *uncertainty quantification* of the bridge as an open problem. Our work builds directly on this sampler: we keep the model frozen and harvest two signals that SelfRDB discards.

2.2 Uncertainty Quantification in Medical Imaging

Per-pixel uncertainty is commonly obtained by MC-Dropout[17], deep ensembles[18], or test-time augmentation[19]; the epistemic/aleatoric taxonomy of [20] frames these as sampling-

based epistemic estimates. Diffusion ensembles follow the same recipe: implicit segmentation ensembles[21] and per-pixel standard deviations across diffusion restorations[22] established that inter-sample dispersion tracks difficult image regions. We inherit this insight but differ in three ways: (i) our ensemble members are sampled from a *diffusion bridge* whose stochastic endpoints and latent variables are native to the sampler, not injected ad hoc; (ii) we pair ensemble dispersion with a second, complementary signal—the convergence residual—that is unique to self-consistent recursive samplers and costs nothing; and (iii) we fuse both into a calibrated trust map with a clinical QC gate rather than reporting dispersion alone. Notably, concurrent bridge models actively *suppress* this stochasticity for reconstruction [23]; we argue it is an asset when the task is trust assessment, even though it is a defect for point synthesis.

2.3 Hallucination Assessment and Trustworthy Radiology AI

Closest to our goal, the Hallucination Index[4] quantifies fabricated content of generative reconstruction models as an *image-level* metric over repeated samples; MUMINS[24] conditions medical image synthesis on metadata and trains an uncertainty-aware prediction head to output a spatial uncertainty map; and the measurement-geometry analysis of [25] proves when a measurement operator can distinguish solutions, explaining hallucination from an operator-design perspective. These approaches either require repeated sampling at image granularity, architectural modification with retraining, or ex-ante operator design. TrustBridge instead derives pixel-level evidence *inside* the existing sampler, requires no architectural change or labels, and additionally supports rule-based gating: our framework is complementary to output-guardrail lifecycles such as RAISE[26] and provides the objective, pixel-level evidence that reporting frameworks like AI-RADS[27] call for, while remaining fully automatic.

Beyond sampling dispersion: semantic and pathology-aware signals.

Dispersion-style evidence is blind to errors the sampler makes *consistently*—a limitation we quantify in §§4. Three concurrent research lines attack this from the evaluation side and mark the natural next step for trust layers like ours. CRAFT[28] scores generated images against clinical criteria with a foundation-model Clinical Alignment Score, reducing hallucination-like content; ViCTr[29] makes synthesis pathology-aware with graded severity control via EWC-regularized LoRA adaptation; and DDWM[30] achieves zero-shot translation by learning only the target-domain distribution and fusing domain-invariant anatomy through wavelet decomposition. All three require either foundation models, retraining, or architectural change; they are complementary to sampler-native evidence and are discussed as future work in §§5.

3. Method

3.1 Preliminaries: the self-consistent recursive diffusion bridge

SelfRDB[3] constructs a bridge whose forward process starts at the target image x_0 and ends at a noise-added source image y_ϵ, with intermediate samples $x_t = \mu_{x_0,t} x_0 + \mu_{y,t} y + \sigma_t \epsilon$ where $\sigma_t^2 = \gamma s_t^2 / (\bar{s}_t^2 + s_t^2)$ grows monotonically toward the source end (*soft prior*). Reverse sampling starts at $x_T = y_\epsilon$; at every step t, a generator $G_\theta(x_t, t, y, \tilde{x}_0^{(r)})$ recursively refines a target estimate $\tilde{x}_0^{(r+1)}$ until the change between successive estimates falls below a threshold τ (SelfRDB: a 1 global mean), after which the self-consistent estimate $\tilde{x}_0^{*}$ drives a posterior draw of x_{t-1}. The publicly released configuration uses $T{=}10$ bridge steps and at most $R{=}2$ recursions, which makes per-slice inference interactive.

3.2 Convergence Residual Field (CReF)

Within one reverse step, the recursion produces a sequence of pixel-wise discrepancies $R^{(r)}(p) = |\tilde{x}_0^{(r+1)}(p) - \tilde{x}_0^{(r)}(p)|$, p indexing pixels. SelfRDB aggregates $R^{(r)}$ to a global mean for its stopping rule and discards the spatial information. We retain the residual field over the final K reverse steps (the steps closest to the data distribution),

$$\mathrm{CReF}(p) \;=\; \frac{1}{K} \sum_{t \in \mathcal{K}} \sum_{r} \big| \tilde{x}_0^{(r+1)}(p) - \tilde{x}_0^{(r)}(p) \big|,$$

where $\mathcal{K}$ is the capture window ($K{=}3$ here). Intuition: pixels the generator keeps revisiting without converging are pixels for which its learned prior provides insufficient evidence—prime candidates for hallucinated content. The signal is epistemic in the sense of[20] and costs nothing: the recursion runs anyway.

3.3 Stochastic-bridge ensemble uncertainty

The bridge endpoint y_ϵ and the generator's internal latent variables z make two runs of the sampler on the same input differ. Running N independent chains (distinct endpoint noise and latents) and taking the per-pixel standard deviation, $U(p) = \mathrm{std}_n\{x_0^{(n)}(p)\}$, yields a second signal that measures *sampling consistency*: pixels whose synthetic value is unstable across chains are regions where the model holds no stable belief. Unlike MC-Dropout[17] or deep ensembles[18], no dropout is injected and no separate networks are trained; the dispersion arises from the bridge sampler itself. Prior diffusion-ensemble work in segmentation and restoration[21,22] established that such dispersion tracks difficult regions; we pair it with a non-dispersion signal (CReF) that is orthogonal to sampling variance.

3.4 Trust map, Trust Score, and QC gate

Both fields are robustly normalized (99th-percentile clipping) and fused as $trustmap(p) = 1 - w_1 \widehat{\mathrm{CReF}}(p) - w_2 \widehat{U}(p)$ with $w_1{=}0.6$, $w_2{=}0.4$. The image-level $TrustScore$ is the body-region mean of $trustmap$ scaled to 0–100, gated by a three-tier rule (PASS/REVIEW/REJECT); §§4 analyzes the operating points actually achieved by these thresholds on three

cohorts. A per-slice report (PNG + JSON) records the maps, the per-step recursion statistics, and the input-gate verdict.

Input gate v2.

Before any synthesis, a six-class classifier over body-standardized percentile and gradient features (brain T1/T2/FLAIR, pelvis T1w/T2w, pelvis CT) verifies that the input matches the declared source modality, with a Mahalanobis out-of-distribution test (threshold at the validation 99.5th percentile) and a flat-image hard rule. In a 1,160-case validation matrix spanning matched, cross-modality, cross-anatomy, and synthetic-OOD suites, *no* illegal input (0/562) received a passing verdict, while no valid input was false-rejected. Mismatched inputs to brain tasks are hard-blocked; mismatched inputs to pelvis-CT tasks are warned and annotated, because unseen-patient pelvic T1w slices are confused with brain classes in 52.5 of cases (§§5).

4. Experiments

4.1 Setup

Datasets.

IXI[31]: 40 healthy subjects (T1/T2/PD), subject-wise split 26/6/8 with no subject overlap; $4,000+$ paired axial slices under the official protocol (per-slice statistics, official mean-norm PSNR/SSIM). **BraTS2018**[32]: 6 HGG patients (60 evaluated tumor-bearing slices) with real co-registered T2 ground truth and tumor segmentation; source FLAIR. **SynthRAD2023 pelvis**[33]: 10 patients, center-matched T1w→CT and T2w→CT, 80 test slices each.

Models.

Two self-trained 128×128 bridges (T2→T1 and T1→T2, 30 epochs, single RTX 4060 Laptop 8 GB, recipe matched to the official release) and two self-trained pelvis bridges (T1w→CT, T2w→CT) provide the quantitative backbone; official SelfRDB weights provide 256×256 demonstration and the BraTS FLAIR→T2 evaluation.

Trust-metric protocol.

Translation quality in Table 1 follows the official whole-image mean-norm protocol for comparability with the literature. *All trust metrics* (AUROC, sparsification, decile separation, correlation) are computed *within a body mask* (source > 0.02): background air is trivially low-error and high-trust, and pooling it with body pixels inflates whole-image trust metrics severalfold. Our own official-weight runs illustrate the effect: the same fused signal scores AUROC 0.78 within the body but 0.90 on the whole image, and its whole-image sparsification gain ($+5.2$ dB) overstates the clinically meaningful within-body gain ($+0.4$ dB) by an order of magnitude. All earlier conclusions in our project that relied on whole-image trust metrics were re-evaluated; two were retracted (§§4.8).

Ethics.

All three datasets are public research datasets; all processing is local; no output is used for clinical decisions.

4.2 The reproducibility crisis

The collapse.

Officially released IXI weights reproduce only ~ 15 dB on independently prepared IXI data—against the paper's 31.56 dB (Table 1; in our runs the official T1→T2 weights average 12.98 dB within the body). A systematic exclusion study over the plausible pre-processing degrees of freedom (rigid misalignment, slice z-shift, intensity windowing, cropping, orientation) traces the gap to pre-processing details the original authors do not publish; the companion SynDiff release is independently reported to have the same problem. We emphasize what this means: *a reader who prepares the data in any reasonable way cannot reproduce the paper*, and the failure mode is invisible—the official outputs are structurally coherent images with a global intensity offset (NCC ~ 0.9), not noise.

What the trust layer does and does not see in a collapse.

A natural hope is that sampler-native signals would "know" the base model is failing. We tested this directly, and the answer is the most important negative result in this paper. On the collapsed official model, the trust map does not fire—mean trust $94.2/100$, low-trust ratio 5.9 (lower than for our healthy self-trained model), all 20 test slices PASS. And the collapse is *not* a benign photometric offset: within-body NCC is 0.41, SSIM 29, and a ground-truth-fitted linear intensity correction leaves PSNR at 12.3 dB—the official model synthesizes *structurally wrong but internally self-consistent* anatomy. The sampler is confident about its wrong output, so CReF and dispersion—which measure self-consistency—are blind, exactly as our tumor analysis predicted at the pixel level, now reproduced at the model level. The lesson for deployment is stark: *training-free internal signals cannot detect a base model that is confidently wrong*. They complement, but do not replace, external validation against ground truth; a guard layer that has never seen its base model fail cannot be trusted to recognize failure.

4.3 Translation quality

Table 1. Main results (official whole-image mean-norm PSNR/SSIM protocol). "Corrected" removes a global linear intensity mismatch whose parameters are estimated from the ground truth; it is an *error decomposition*, not a method metric—the raw column is the honest operating point. IXI: self-trained 128×128. Pelvis: self-trained, 80 test slices each. BraTS: official FLAIR→T2 weights, 60 tumor-bearing slices.

Dataset	Direction	PSNR (dB)	SSIM (%)	NCC	Corrected PSNR
IXI (paper value[3])	T2→T1	31.56	95.65	–	–
IXI, ours	T2→T1	19.81±1.10	71.0±6.6	0.884	31.74

IXI, ours	T1→T2	20.44±1.16	80.4±4.1	0.905	32.52
Pelvis, ours	T1w→CT	24.36±2.92	81.6±4.8	0.962	–
Pelvis, official	T1w→CT	21.24±2.42	64.3±6.0	0.936	–
Pelvis, ours	T2w→CT	23.13±0.83	74.5±2.7	0.961	–
Pelvis, official	T2w→CT	19.04±0.73	57.3±2.9	0.893	–
BraTS2018, official	FLAIR→T2	24.06±1.62	87.4	0.944	–

Table 1 reports translation quality. On IXI, decomposing the error shows that the content fidelity of our 128×128 self-trained models matches the paper's setting: after removing a global linear intensity mismatch, PSNR reaches $31.74/32.52$ dB (T2→T1/T1→T2) versus the paper's 31.56 dB, while the raw gap is attributable to that offset and the reduced training budget; we emphasize that the corrected column uses ground-truth-estimated parameters and serves only as a diagnosis. Our self-trained pelvis bridges beat the official weights by $+3.1/+4.1$ dB with real CT ground truth. On tumor patients, the official FLAIR→T2 weights reach 24.1 dB (whole-image protocol; 18.3 dB within the body) against real co-registered T2—a quantitative validation of bridge translation on *pathological* data in our pipeline.

4.4 Trust localization, UQ baselines, and the failure boundary

Table 2. Uncertainty-quality comparison under the within-body protocol (IXI test, 60 slices per direction, seed 2026). **Top:** self-trained 128×128 models (the quantitative backbone); all dispersion baselines use the same forward-pass budget as our ensemble (40 generator calls); CReF (N=1) uses a single chain (10 calls, *zero extra cost*). Sep is the median-across-slices ratio of mean error between least- and most-trusted deciles; for the self-trained T1→T2 model the most-trusted decile carries near-zero error, so Sep is an order-of-magnitude statement rather than a precise value. **Bottom:** the same protocol on the official weights, where the signal ranking changes and T2→T1 fails entirely.

	T1→T2 (self)			**T2→T1 (self)**		
Signal	AUROC	ΔPSNR@20%	Sep	AUROC	ΔPSNR@20%	Sep
Random (control)	0.498	≈0	1.0	0.501	≈0	1.0
Gradient	0.225	−0.45	0.54	0.475	−0.55	0.40
Input-noise dispersion	0.861	+1.47	65	0.739	+0.97	3.0
TTA dispersion	0.820	+1.33	6.3	0.711	+0.79	1.9
CReF only (N=1, free)	0.880	+2.28	103	0.792	+1.51	2.8
Ensemble only (N=4)	0.897	+2.34	159	0.806	+1.63	3.8
Fused (ours)	**0.900**	**+2.67**	**182**	**0.816**	**+1.85**	**3.7**
Same protocol on the official weights (N=4 fused):						
Fused (official)	0.783	+0.37	–	0.465	+0.21	–
TTA (official)	**0.808**	+0.70	–	0.482	+0.95	–
All signals (official, T2→T1)	–	–	–	0.465–0.505 (chance)		

Table 2 carries two messages. **On self-trained models**, our fused signal wins outright: AUROC $0.900/0.816$ (T1→T2/T2→T1) with $+2.67/+1.85$ dB sparsification gains, ahead of every equal-budget baseline, and a single CReF chain (zero extra cost) already reaches $0.880/0.792$—most of the value at a quarter of the budget. Fusion is robust but does not beat the ensemble alone by much; CReF's claim is cost, not accuracy. **On the official weights**, the ranking changes and the T2→T1 direction collapses entirely: every dispersion signal—ours and all baselines'—is at chance (0.465–0.505), because that model's residual error is dominated by fine texture that no dispersion measure can track. *The same direction, the same protocol, opposite conclusions on two models*: the failure boundary is a property of the (model, direction) pair, not of the signals, which is why we recommend validating uncertainty signals per (model, direction) before any deployment (§§5). A CReF-triggered routing scheme (uncertain regions re-examined by TTA) was tested to recover full-budget TTA quality at lower cost; on the self-trained model the curve is flat—pure CReF (0.880 at budget 10) already dominates full TTA (0.820 at budget 50)—so routing adds nothing there, a further confirmation that which signal wins is model-dependent.

4.5 Calibration

Within the body, the least-trusted decile carries 1.9–$2.6\times$ the mean absolute error of the most-trusted decile on T1→T2 (Table 2, Sep column) and $2.1\times$ on BraTS. These separations are far more modest than whole-image numbers suggest (35–$73\times$), because whole-image deciles are dominated by the air/body contrast. The trust map is best read as a *ranking within the body*, not an absolute error magnitude.

4.6 Image-level QC operating points

The image level is where the layer is weakest, and we quantify this rather than hide it. Across three cohorts (IXI both directions, 60 slices each; BraTS, 60 tumor slices), the slice-level Trust Score (body mean, 0–100) correlates with slice quality (body PSNR) weakly: Pearson $r = +0.47$ (T1→T2, self-trained), $r = +0.26$ (T2→T1, self-trained), $r = -0.10$ (BraTS, official)—right-signed on the models whose signals work, absent on the one whose do not. The factory thresholds are likewise mis-calibrated to the actual score distributions: PASS≥75 flags 88 of pooled slices while REJECT<55 fires on 3. An empirical reading: the lowest-scoring slices are enriched for quality problems but coverage–precision trade-offs are steep (at the best pooled cut, 9 coverage buys 29 precision for the bottom-quality quartile). We therefore draw a stronger conclusion than “the thresholds need re-tuning”: *global scalar trust scores should not be used as QC gates for medical image translation at all*. Slices differ so much in anatomy that no single body-mean score can separate good from bad across a cohort. We recommend the community standardize on pixel-/region-level trust evidence (the trust map, low-trust region lists) for human review, and treat any image-level gate—ours included—as a triage heuristic at best.

4.7 Boundary experiments

Seven controlled experiments delineate what the layer can and cannot do at this data scale: (1) *learnable aggregation*: a 16-feature logistic/GBDT aggregator trained on the validation split improves T2→T1 marginally ($+0.010$ AUROC) but regresses T1→T2 (-0.03), overfitting the 6-subject validation set; (2) *image-level regression*: a 16-feature ridge model predicts image PSNR worse than the baseline trust mean (0.32–0.33 vs. 0.38–0.40); (3) *adaptive resampling* and (4) *round-trip consistency signal*: no quality gain (-0.016 dB and AUROC 0.503 respectively) at 35–75 extra cost—consistent with ensemble-mean saturation; (5) *intensity harmonization*: post-hoc quantile/affine matching degrades raw PSNR (-0.8/-0.9 dB) because the per-slice intensity offset is not predictable without ground truth; and (6) *T2→T1 signal failure on the official weights* (§§4.4): all dispersion signals at chance when that model's residual error is texture-dominated—while the same direction works on our self-trained model (0.816), isolating the failure to the model, not the task; and (7) *semantic-proxy signals do not complement dispersion*: block-wise normalized mutual information and gradient-orientation agreement between source and synthesis reach only AUROC 0.42–0.67 where dispersion reaches 0.82–0.90, and their combination with dispersion does not improve it—structural agreement with the *source* is a weaker error surrogate than sampler-native dispersion. These results are reported in full as negative findings: at 40 subjects, the pixel-wise trust map is the only reliable product form, and learning-based aggregation requires ≥20 validation subjects.

4.8 Tumor-region analysis

On the BraTS cohort (official FLAIR→T2 weights, 60 tumor-bearing slices, within-body protocol): (i) pixel error inside the tumor is $1.60\times$ the error outside (90 of slices), confirming that synthesis risk concentrates at the lesion; (ii) the trust map localizes the top-error decile with AUROC 0.911 ± 0.037 and a $+3.12$ dB sparsification gain at 20%—trust signals survive the protocol correction on this model; but (iii) we *retract* an earlier claim: low trust does *not* localize tumor tissue itself (within-body AUROC vs. segmentation 0.51 ± 0.10; our previously reported 0.871 was a whole-image artifact of air negatives, and mean trust is in fact slightly *higher* inside tumors, 0.772 vs. 0.727 outside). Flagged regions indicate where the model is unsure about its synthesis, not where pathology is. The lesson goes beyond our method: pixel-variance-style UQ measures the *variance of the synthesis process*, not the *clinical salience* of the content—a model can be highly confident (and highly consistent) about a pathologically wrong synthesis. Any UQ layer for clinical use must be audited against pathology annotations, not only against error maps.

4.9 Stress tests: input degradation and cross-institution transfer

Input degradation.

We degrade the input along three axes—Gaussian blur ($\sigma \in \{0.5, 1, 2\}$), Gaussian noise ($\sigma \in \{0.01, 0.03, 0.06, 0.1\}$), and horizontal misalignment (2–8 px)—over 30 IXI test slices per dose. Translation quality degrades mildly (body PSNR $12.9 \rightarrow 11.8$ dB at the worst dose). The response of our trust machinery is asymmetric and instructive: the *input gate's* Mahalanobis distance fires dose-dependently for blur ($2.9 \rightarrow 15.0$) and noise ($\rightarrow 9.3$) but is blind to small misalignment ($\rightarrow 2.9$); the *trust map* does not flag degraded inputs at all—moderate blur actually *raises* mean trust ($0.63 \rightarrow 0.70$), because a smoother input is easier for the sampler to be self-consistent about—and the within-body localization AUROC decays with dose ($0.79 \rightarrow 0.70$ under noise; $\rightarrow 0.67$ under misalignment). Knowing-what-you-don't-know here is carried by the *input gate*, not the trust map; misalignment detection remains open (we discuss this in §§5).

Cross-institution transfer.

The strongest external objection to sampler-native trust evidence is that it may be locked to the training cohort. We therefore evaluate the identical pipeline (official FLAIR→T2 weights, identical protocol, never-seen subjects) on two held-out sets from BraTS2018 raw data: six patients from the same collection as our reference cohort (patient-level generalization) and six `CBICA` validation subjects prepared by a different institution (institution-level generalization), 120 slices each. Trust localization *transfers*: AUROC 0.863 ± 0.155 (held-out patients) and 0.950 ± 0.021 (cross-institution) versus 0.911 ± 0.037 on the reference cohort, with sparsification gains $+2.78$ and $+3.12$ dB respectively. The caveats are real—same preprocessing recipe on our side, six subjects per set—but the direction is clear: within this cohort family, the signals are not an artifact of having seen the patients.

5. Discussion

Reproducibility.

Officially released IXI weights reproduce only ∼15 dB on independently prepared IXI data (the paper reports 31.56 dB). A systematic exclusion study (rigid misalignment, z-shift, intensity window, cropping, orientation) traces the gap to preprocessing details that the original authors do not publish; the same issue is independently reported for the companion SynDiff release. We therefore trained same-recipe models on our own fully documented pipeline and base all quantitative claims on them, while using the official weights for high-resolution demonstration.

Quality and trust calibration are separable.

On BraTS, our self-trained models beat the official weights on PSNR by $+3$–4 dB, yet the official weights' trust maps localize error far better. The T2→T1 result sharpens the point: a model can be accurate and still produce uninformative uncertainty, when its residual error lacks the spatial structure that dispersion measures respond to. Uncertainty signals must be

validated per (model, direction) pair before deployment—a conclusion our own negative result makes concrete.

The protocol trap.

Whole-image evaluation of pixel-wise trust metrics is systematically optimistic whenever images have trusted, low-error background. Because this describes nearly all medical image translation settings, we recommend body-masked evaluation as standard practice for uncertainty quantification in this field, and we publish our retraction trail (docs/18 and the corrected analysis scripts) as evidence that this is easy to get wrong—we did.

Limitations.

(1) The trust map flags *unreliable* regions, not *wrong* ones: a confidently synthesized hallucination is invisible to it—and the blindness extends to the model level: a collapsed base model that is confidently wrong receives PASS ($94/100$ mean trust, §§4.2). (2) Learnable aggregation overfits 6-subject validation sets; reopening the learnable route requires ≥20 validation subjects. (3) The input gate hard-blocks brain-task mismatches but only warns on pelvis CT tasks: unseen-patient pelvic T1w is confused with brain classes in 52.5 of cases, so hard blocking would reject half of the valid inputs; the Mahalanobis OOD threshold is additionally inflated by cross-scanner domain shift, leaving some structured synthetic probes to the classifier layer rather than the distance test. (4) Image-level QC and tumor localization remain unsolved at this scale; the trust map additionally does not respond to input degradation—blur even raises it (§§4.9)—so degraded-input detection relies on the input gate, and small misalignments evade both. (5) The three-tier gate thresholds require per-deployment re-tuning (§§4). (6) All outputs are research demonstrations and must not guide clinical decisions.

6. Conclusion

We turned the discarded internals of a state-of-the-art diffusion-bridge sampler into a pixel-wise trust map, score, and QC workflow for medical image translation, and subjected it to a harsher evaluation than we originally planned: body-masked metrics, equal-budget dispersion baselines, and per-cohort operating-point analysis. What survives is a clear, bounded picture: on models whose errors are spatially structured (T1→T2; official weights on tumor patients), the training-free signals localize unreliable regions well (AUROC $0.82–0.95$) and CReF supplies most of that value at zero cost; on texture-dominated error the signals—and every baseline we tried—are blind. The broader messages are methodological: generative samplers compute more evidence than they use, and harvesting it converts a synthesis engine into a trust-assessable system; but uncertainty claims in image translation must be evaluated within the body and against cheap baselines, or they will be overstated. Future work will reopen learnable aggregation at larger subject scale, extend the analysis to 256×256 models, and integrate radiologist feedback.

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References

1. Ho et al. (2020). Denoising Diffusion Probabilistic Models. *Advances in Neural Information Processing Systems*.
2. Nichol et al. (2021). Improved Denoising Diffusion Probabilistic Models. *International Conference on Machine Learning*.
3. Arslan et al. (2025). Self-Consistent Recursive Diffusion Bridge for Medical Image Translation. *Medical Image Analysis*.
4. Tivnan et al. (2024). Hallucination Index: An Image Quality Metric for Generative Reconstruction Models. *Medical Image Computing and Computer Assisted Intervention (MICCAI)*.
5. Zhu et al. (2017). Unpaired Image-to-Image Translation Using Cycle-Consistent Adversarial Networks. *IEEE International Conference on Computer Vision*.
6. Isola et al. (2017). Image-to-Image Translation with Conditional Adversarial Networks. *IEEE Conference on Computer Vision and Pattern Recognition*.
7. Dar et al. (2019). Image Synthesis in Multi-Contrast MRI with Conditional Generative Adversarial Networks. *IEEE Transactions on Medical Imaging*.
8. Dalmaz et al. (2022). ResViT: Residual Vision Transformers for Multi-Modal Medical Image Synthesis. *IEEE Transactions on Medical Imaging*.
9. Özbey et al. (2023). Unsupervised Medical Image Translation with Adversarial Diffusion Models. *IEEE Transactions on Medical Imaging*.
10. Korkmaz et al. (2025). I2I-Galip: Unsupervised Medical Image Translation Using Generative Adversarial CLIP. *Medical Imaging with Deep Learning (MIDL)*.
11. Song et al. (2020). Score-Based Generative Modeling through Stochastic Differential Equations. *arXiv:2011.13456*.
12. Song et al. (2019). Generative Modeling by Estimating Gradients of the Data Distribution. *Advances in Neural Information Processing Systems*.
13. Chung et al. (2023). Direct Diffusion Bridge Using Data Consistency for Inverse Problems. *arXiv:2305.19809*.
14. Su et al. (2023). Dual Diffusion Implicit Bridges for Image-to-Image Translation. *arXiv:2203.08382*.
15. Kim et al. (2024). Unpaired Image-to-Image Translation via Neural Schrödinger Bridge. *International Conference on Learning Representations*.
16. Delbracio et al. (2023). Inversion by Direct Iteration: An Alternative to Denoising Diffusion for Image Restoration. *Transactions on Machine Learning Research*.
17. Gal et al. (2016). Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. *International Conference on Machine Learning*.
18. Lakshminarayanan et al. (2017). Simple and Scalable Predictive Uncertainty Estimation Using Deep Ensembles. *Advances in Neural Information Processing Systems*.
19. Ayhan et al. (2018). Test-Time Data Augmentation for Estimation of Heteroscedastic Aleatoric Uncertainty in Deep Neural Networks. *Medical Imaging with Deep Learning (MIDL)*.
20. Kendall et al. (2017). What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision? *Advances in Neural Information Processing Systems*.

21. Wolleb et al. (2022). Diffusion Models for Implicit Image Segmentation Ensembles. *Medical Imaging with Deep Learning (MIDL)*.
22. Osuna-Vargas et al. (2024). Denoising Diffusion Models for High-Resolution Microscopy Image Restoration. *arXiv:2409.12078*.
23. Song et al. (2024). Self-Consistent Nested Diffusion Bridge for Accelerated MRI Reconstruction. *arXiv:2412.09998*.
24. Oliveras et al. (2026). MUMINS: Metadata-Conditioned Uncertainty-Aware Medical Image Next-State Synthesis. *arXiv:2609.17169*.
25. Jin et al. (2026). Measurement Geometry and Design for Trustworthy Generative Inverse Problems. *arXiv:2606.02309*.
26. Cardoso et al. (2023). RAISE – Radiology AI Safety, an End-to-End Lifecycle Approach. *arXiv:2311.14570*.
27. Fink et al. (2026). Artificial Intelligence Reporting and Data System (AI-RADS): A Structured Framework for Case-Level Assessment of AI Output. *Investigative Radiology*.
28. Chung et al. (2026). CRAFT: Clinical Reward-Aligned Finetuning for Medical Image Synthesis. *arXiv preprint arXiv:2605.12650*.
29. Susladkar et al. (2025). ViCTr: Vital Consistency Transfer for Pathology-Aware Image Synthesis. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.
30. Li et al. (2025). Denoising Diffusion Wavelet Models for Zero-shot Medical Image Translation. *Knowledge-Based Systems*.
31. ixi_dataset (2008). IXI Dataset.
32. Menze et al. (2015). The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS). *IEEE Transactions on Medical Imaging*.
33. Thummerer et al. (2023). SynthRAD2023 Grand Challenge Dataset: Generating Synthetic CT for Radiotherapy. *Medical Physics*.